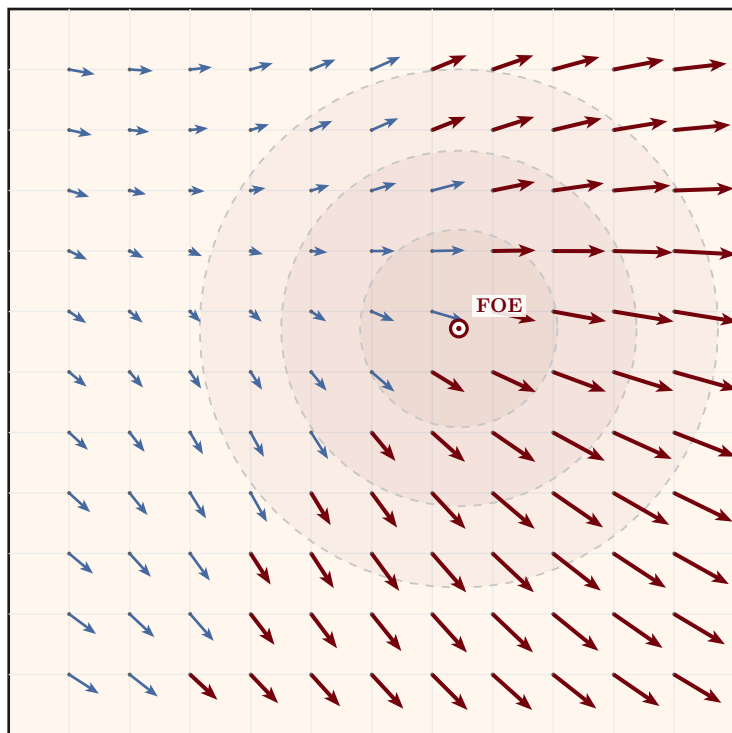


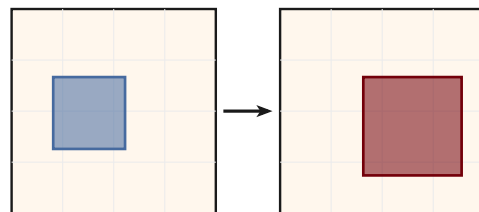
# Optical flow field

dense motion  $\mathbf{v}(\mathbf{x}) = (u, v)$  from  $I_t \rightarrow I_{t+1}$ , one arrow per pixel grid cell



$$\text{brightness constancy } I_{t(x)} = I_{t+1}(x + v) \rightarrow I_x u + I_y v + I_t = 0$$

## input frames

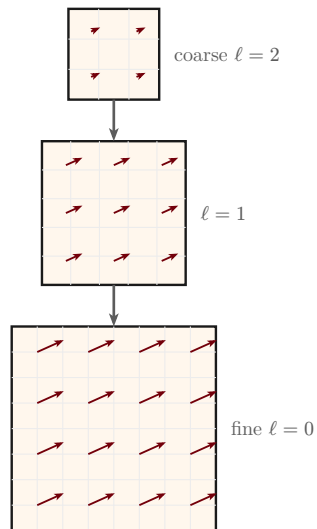


$I_t$

$I_{t+1}$

*displacement of the patch  $\rightarrow$  flow  $\mathbf{v}$*

## coarse-to-fine warping



*upsample  $\times 2$  + warp  $\rightarrow$  refine*

- fast — object (looming)
- slow — background pan
- ⊙ focus of expansion (FOE)